Reyshwanth Ganeshan

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Education

TEXAS A&M UNIVERSITY 2023–2027
Ph.D. Mech. Eng., GPA: 3.5*/4.0
Advisor: Dr. Sivakumar Rathinam

TEXAS A&M UNIVERSITY 2021–2023
M.S. (Thesis) Mech. Eng., GPA: 4.0/4.0

IIT MADRAS 2017–2021
B.Tech. Mechanical Engineering

Skills

Programming: C++, Python, ROS/ROS2, OpenCV, Julia, Docker, Git
ML: Pytorch, Tensorflow, CUDA, SQL, Stable-Baselines3, Gymnasium
Tools: Gurobi, CPLEX, PostgreSQL, QGIS, Gazebo, Unity, IsaacSim, MATLAB

Selected Courses

Convex Opt., **Deep RL**, **ML**
Autonomous Decision Making
Intelligent Transportation Systems
Algorithms, Nonlinear Control
Control Design, Statistics

Publications

- **R. Ganeshan**, et al. Cooperative **MAPPO**. *Submitted to ICRA*
- N. Kumar, **R. Ganeshan**, et al. (2026). Privacy-Preserving Cooperative Localization. *IROS 2026*
- **R. Ganeshan**, et al. (2026). Bi-Objective Dynamic Zone Navigation. *AIAA SciTech 2026*
- N. Kumar, **R. Ganeshan**, et al. (2026). UAS Fleet Planning in GPS-Denied Environments. *AIAA SciTech 2026*
- **R. Ganeshan**, et al. (2025). Multi-Domain UGV-UAV Path Planning. *AIAA SciTech 2025*
- **R. Ganeshan**, S. Darbha (2024). Diagnostic ML for Vehicle Air Brakes. *IFAC*

Achievements

- Led a 35-person **AV** team to two straight **2nd-place** Autodrive finishes (2024–25)
- Led the team to Autodrive overall **2nd** among 10 teams across 5 years
- Co-led a 45-person **FSAE** team; Formula Bharat 2020 National Champions
- Top **0.16%** in **IIT-JEE Advanced 2017** among **1.2M** candidates

Professional Experience

TEAM CAPTAIN Aug 2023 – June 2026
GM SAE Autodrive Challenge II
- Scored constraint-aware **MPC** trajectories on rasterized costmaps, using that safety metric to compare obstacle avoidance before on-vehicle runs
- Set up autonomous vehicle behavior with behavior trees on the **Nav2** stack
- Deployed **L4** vehicle with multi-sensor fusion for robust pose estimation
- Built **LiDAR SLAM** with probabilistic filtering for GPS-denied urban localization
- **PointPillar** + **YOLOv8** multi-class detection at **80 ms** (Deer/Barrel/Barricade)
- Multi-modal **LiDAR/Radar/camera** tracking for fog, glare, and rainy conditions

VEHICLE SOFTWARE INTERN Aug 2022 – Dec 2022
Tesla Inc.
- Built **HIL Firmware Regression Testers** for Tesla vehicles (S/3/X/Y and more)
- Python/**CAN** testing for safety-critical maneuvers (ABS); **60%** faster testing

GRADUATE RESEARCH ASSISTANT Aug 2023 – Present
Autonomous Systems Lab
- Evaluated **DeepRL** against A*/Dijkstra for GPS-denied routing, outdoor demos
- Developed multi-agent navigation algorithms with outdoor UAV-UGV demos
- Applied research in **robotics, AI, and multi-agent navigation**

RESEARCH INTERN, LOCALIZATION & MAPPING Aug 2026 – Nov 2026
Plus AI
- **LiDAR localization**: runtime gains; fusion under uncertainty/adverse conditions
- **Automapping** pipeline to flag remaps and manage 3D LiDAR map health
- Learning-based **BEV** model to auto-update HD maps on road geometry change

Selected Projects

BI-OBJECTIVE DYNAMIC ZONE AVOIDANCE Jan 2025 – June 2025
Texas A&M University
- Designed multi-objective A* risk metrics that turn UAV danger into a numeric cost, traded off against traversal time in shape-changing zones
- **BOA*** planning for dynamic zone navigation on fixed-wing aircraft
- Proximity-based risk scoring; **72%** faster convergence via better exploration

UNCERTAINTY-GATED COVERAGE (GPS-DENIED) Sept 2025 – Present
Submitted to ICRA
- Evaluated multi-agent **DeepRL** vs RH-MPC (**83%** vs **46%**) on gated coverage
- Added cooperative **EKF** ranging; ablating it cuts high-drift coverage **96%** to **69%**
- Built **7**×**7** egocentric crops enabling **zero-shot** transfer to 30×30 maps (∼**99.6%**)

MULTI-DOMAIN UAV-UGV TEAM Aug 2024 – Dec 2024
Texas A&M University
- Deployed UAV-UGV team with **GPLA*** for search and rescue
- Costmap avoidance and replanning on Clearpath Huskys (LiDAR)
- **Gazebo** validation with UR5 clearing and online graph replanning

PRIVACY-PRESERVING RANGE (GPS-DENIED) Oct 2025 – Jan 2026
Texas A&M University
- Built **privacy-preserving SDP/MVE** localization with dual-only exchange
- Derived convex range-pair constraints that tighten sets beyond spherical bounds
- Deployed **UWB-EKF** localization on UAV-UGV (2 drones + Husky; 300×150 ft)